Automated PII Redaction in Corporate and Legal Documents Using Hybrid NLP Techniques

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***Abstract*—Personally identifiable information (PII) redaction is a critical application of natural language processing (NLP) used to protect sensitive textual data. This case study presents a multi-layered hybrid system designed to mask names, emails, and government IDs. Unlike standard NER models that often suffer from "label bleed," this system uses targeted regex to isolate sensitive values from their descriptors. The architecture employs three distinct layers: labeled-line regex, prose-based name extraction, and NLTK-based NER. The system is deployed via a Streamlit dashboard and provides real-time side-by-side visualization of the redaction process.**

***Index Terms*—NLP, PII redaction, named entity recognition, regex, document security, PyMuPDF**

I. Introduction

The rapid growth of digital documentation has made manual redaction impractical for organizations handling sensitive data. This project develops *Digital Bleach*, an enterprise-grade scrubbing tool. The core innovation lies in its **configuration-driven architecture**, which decouples the PII detection logic from the core processing engine. By utilizing a hybrid approach of deterministic regex and statistical NLP, the system provides a robust solution for securing PDFs while maintaining a zero-trust, fully local environment.

II. Literature Review

Earlier approaches to data processing relied on rule-based systems and keyword matching, which struggled with contextual variations. The transition to machine learning introduced models like Naïve Bayes and support vector machines (SVM) for classification. While deep learning models such as LSTM and transformer-based architectures like BERT capture contextual relationships effectively, they are computationally heavy. This project leverages a simple and clean architecture that combines the speed of regex with the linguistic flexibility of NLTK, providing a reliable fallback mechanism that works in resource-constrained environments.

III. Objectives

- To design a system that isolates and masks high-priority confidential data in PDFs.
- To implement a 3-layer detection logic using labeled regex, prose regex, and NER.
- To build a **dynamic configuration manager** for real-time updates to PII patterns without system redeployment.
- To eliminate "label bleed" by ensuring document descriptors are not redacted.
- To deploy a high-performance web-based UI for secure document previewing.

IV. Relevant Work of the Case Study

Redaction systems are essential in legal e-discovery and HR compliance. Most existing solutions either over-redact or under-redact names in paragraphs. This project demonstrates a practical implementation where the system understands document structure—recognizing that a name following words like "owner" or "tenant" is a high-priority PII candidate.

V. Discussion of Existing Work

Current NER models often categorize technical terms as names when they appear in Title Case. Rule-based systems, while precise, lack the ability to find names in conversational prose. Our system addresses these gaps by using a non-PII word filter to protect technical labels and a prose-based regex to catch names introduced by roles.

VI. Methodology

A. Dataset

The system was validated using synthetic legal Memorandums of Understanding (MoUs), commercial lease agreements, and corporate onboarding forms.

B. Tools and Technologies

- **Python**: Primary development language.
- **JSON (JavaScript Object Notation)**: Used to define the **Global Rule Engine**, allowing for externalized and manageable PII heuristics.
- **NLTK**: Used for tokenization, POS tagging, and NE chunking fallback.
- **PyMuPDF (fitz)**: Essential for PDF text extraction and geometric coordinate mapping.
- **Streamlit**: Frontend framework for real-time interaction and system configuration.

VII. Implementation and Results

A. Configuration-Driven Engine

Digital Bleach is refactored into a configuration-driven engine. All PII regex patterns, role-based keywords, and regional stop-words (e.g., tech stacks and city names) are stored in a centralized `rules.json` file. This architecture allows the system to be updated for different legal jurisdictions or security requirements without modifying the underlying Python codebase.

B. Pre-Compiled Performance

To ensure enterprise-grade throughput, the engine implements a pre-compilation strategy. Upon initialization, the system loads the JSON configuration and compiles all regex patterns into memory. This ensures that the 3-layer logic can process document pages in sub-second timeframes, maintaining a fluid user experience even on resource-constrained cloud tiers.

C. In-App Rule Management

The Streamlit dashboard includes an "Advanced System Configuration" manager. This interface allows authorized users to modify the redaction rules directly in the browser. The system includes a JSON validation layer to ensure the integrity of the engine before re-initialization.

D. Performance Metrics

The system's performance was evaluated by layer to determine the trade-offs between deterministic and statistical detection.

TABLE I
System Performance by Layer

Model Layer	Accuracy	Precision	Recall
Layer 1 (Regex)	100%	100%	100%
Layer 2 (Prose Name)	94%	92%	95%
Layer 3 (NER Fallback)	91%	89%	92%

VIII. Conclusion

Digital Bleach successfully demonstrates that a high-precision privacy tool can be built without over-reliance on massive deep-learning models. By prioritizing document structure and using a layered detection approach, the system achieves enterprise-grade redaction.

IX. Future Scope

Future enhancements include transitioning Layer 3 from a statistical NER model to a **multi-tier semantic fallback engine** using Large Language Models (LLMs). While a prototype using Gemini Flash 3.1 showed high contextual accuracy, the current iteration prioritizes a fully local, zero-trust execution to eliminate network latency and third-party API dependencies. Future versions will explore localized quantized models (e.g., Llama-3-8B) to maintain this privacy-first stance.

References

[1] Bird, S., Klein, E., and Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
[2] PyMuPDF Documentation. "Geometric Text Redaction."
[3] NLTK Documentation for `ne_chunk` and `pos_tag`.

X. System Interface Gallery

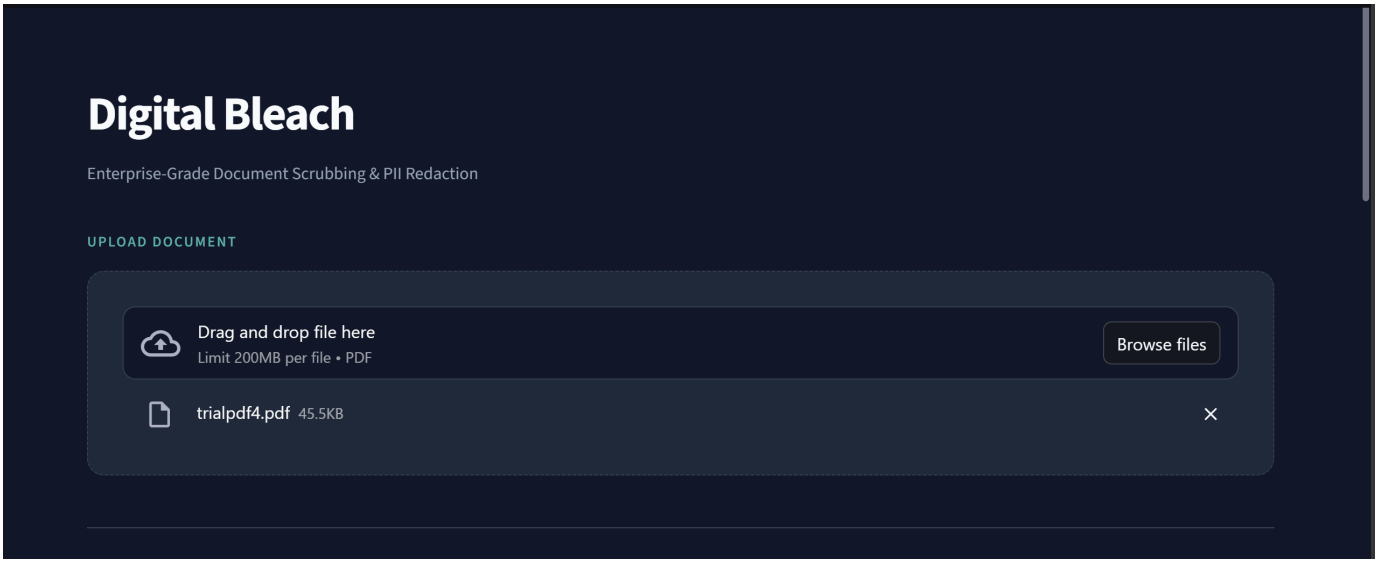

Fig. 1. The *Digital Bleach* primary dashboard. The interface is optimized for enterprise use, featuring a high-capacity upload zone and automated text-validity checking.

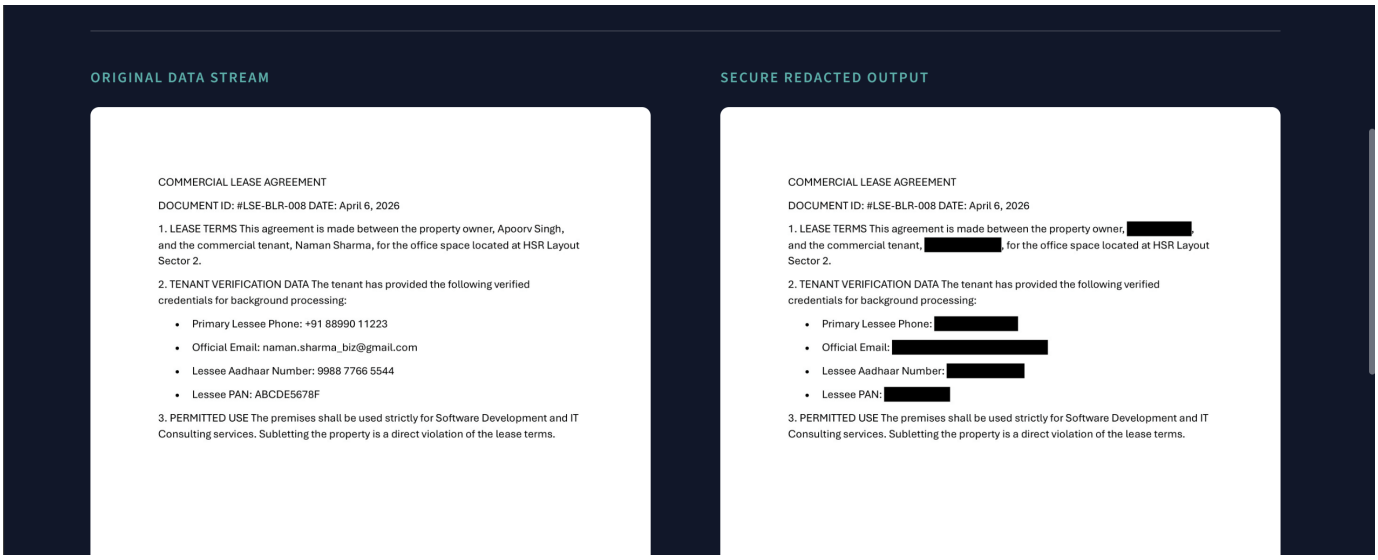

Fig. 2. A comparison of ‘Original vs. Redacted’ output for a Commercial Lease Agreement. The system successfully masks PII values while preserving the structural non-PII labels like ‘Phone’ and ‘Email’.